English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, 16.12.2023 Test 1

The exact grading formula depends on the results. As an approximation, use 2 plus the points earned. Time allowed: 3 hours. Good luck.

m, n, k denote nonnegative integers.

Problem 1. 1. (0.25 points) An urn contains 4 yellow, 5 green and 2 blue pebbles. After drawing one, a person rolls fair dice: one die for yellow, two for green and three for blue. Given that the dice total is 3, what is the probability that the pebble was green?

2. (0.25 points) A fair die is rolled five times. Find the probabilities of
$A = \{$the maximum value is at least 3$\}$ and
$B = \{$the product of all values is divisible by 10$\}$.

3. (0.25 points) A person tosses two biased coins, each with heads probability 1/3. If their faces agree, they draw two balls from an urn containing 4 black and 2 white balls, winning as many BGN as the number of white balls drawn. If the coin faces differ, they roll a die and pay BGN 3 if it is even. Find the distribution of the profit and prove that the game is unfair. How much should the player pay for the game to be fair?

Problem 2. 1. (0.25 points) Let $X \sim Ge(p)$, i.e. $\mathbb{P}(X = k) = pq^k$ for $k \geq 0$; Prove the memoryless property: for every $m, n \geq 0$, $\mathbb{P}(X > m + n | X \geq n) = \mathbb{P}(X > m)$.

2. (0.25 points) Prove the converse: if Y is a random variable taking values in $\{0, 1, 2, \dots\}$ and, for all $m, n \geq 0$, $\mathbb{P}(Y > m + n | Y \geq n) = \mathbb{P}(Y > m)$, then $Y \sim Ge(p)$ for some $p \geq 0$.

3. (0.25 points) Let $X_1, \dots, X_n$ be independent random variables and define $X_{\min} := \min\{X_1, \dots, X_n\}$ and analogously $X_{\max}$. Which expressions

$$\mathbb{P}(X_1 > m)^n,\ \mathbb{P}(X_1 \leq m)^n,\ \mathbb{P}(X_1 \leq m),\ 1 - m\mathbb{P}(X_1 > m)^n,\ 1 - \mathbb{P}(X_1 > m)^n, (1 - m)(1 - \mathbb{P}(X_1 > m)^n)$$

give the probabilities respectively $\mathbb{P}(X_{\min} \leq m)$ and $\mathbb{P}(X_{\max} \leq m)$? Prove your answer.

4. (0.25 points) Five fair dice are rolled together. Any die showing six is removed, and the others are rolled again until none remain. Let K be the total number of rounds. Find $\mathbb{P}(K \leq n)$ and $\mathbb{P}(K = n)$.

Problem 3. Genetic algorithms are a well-known class of heuristic algorithms. They begin with an initial population of chromosomes encoding possible solutions to a problem; over time, it evolves into a better population, hence their name.

Specifically, suppose we wish to initialize n chromosomes (Boolean vectors) of length m. All genes (vector entries/bits) are independent $Ber(1/2)$ random variables.

1. (0.5 points) What is the probability that all vectors are distinct? What is the expected total number of ones in all vectors? What is the expected number of vectors with fewer than $m/4$ zeros?

2. (0.5 points) What is the expected number of repetitions, i.e. equal pairs of vectors among the m vectors? Answer the same question if the vector entries are independent $Ber(p)$ with $p \in (0, 1)$. For which p is the expected number of repetitions minimized?

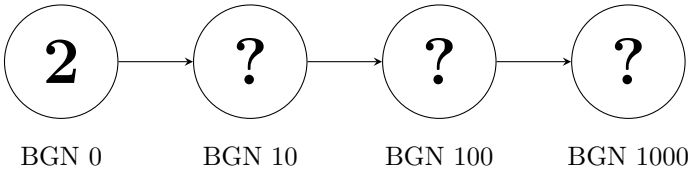
3. (0.5 points) **Bonus: What is the expected number of unique vectors*?

4. (0.5 points) ***Bonus: For fixed m, what is the order of the smallest n for which the probability that all chromosomes are distinct is below 50%?*

Problem 4. A television lottery uses a random generator that selects an integer uniformly from 1 to 5. The game is as follows[1]:

- A random number is selected (2 in the example);
- The player may stop and win the amount below the last number drawn, or guess whether the next number will be smaller or larger. **Consecutive numbers are guaranteed to differ**;
- An incorrect guess wins nothing and ends the game. After a correct guess, the player may continue, up to the grand prize of BGN 1000.

[1] https://shorturl.at/bjlqU



1. (0.5 points) Propose a strategy and calculate the winning probability and expected payoff if the player has decided to make exactly k guesses, for $k = 0, 1, 2, 3$. Compare with independently choosing larger/smaller at random with probability $1/2$ each time.

2. *(0.25 points) Bonus: Give pseudocode solving the problem for numbers between* 0 *and* n*,* m *player choices, and prizes* $c_0, \ldots, c_n$*.*

3. *(0.5 points) Bonus: Give pseudocode deciding whether stopping or guessing again is optimal after each round in the general game of part 2.*